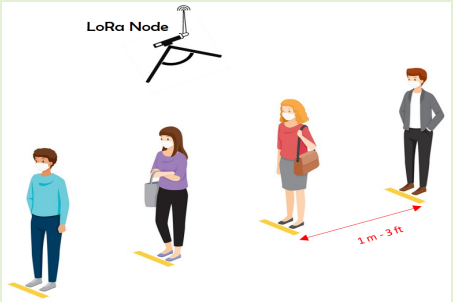
Monitoring Social Distancing in Queues Using Infrared Array Sensor

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***Abstract*—Since the emergence of the coronavirus disease (COVID-19), more than 165 million people have been infected all around the world (as of May 2021). The COVID-19 virus is highly contagious, especially in large crowded spaces. To prevent the infection and slow the transmission of the virus, the World Health Organization recommends wearing a mask, washing hands regularly, and maintaining a distance of at least one meter between people. To support this prevention, we propose a connected object based on LoRa technology to monitor the social distancing of one meter between people in queues. The proposed system uses infrared array sensor MLX60940 to detect people through their body temperature. Based on this output, we developed an algorithm to verify social distancing. The main advantage of this node is its ability to preserve the privacy of detected people and consume less energy compared to camera-based systems. Experimental results show that the proposed node can detect up to six persons per queue with an accuracy of 93% when the distance between people is below 80 cm.**



***Index Terms*—Social distancing, infrared array sensor, LoRa technology, low power.**

I. Introduction

THE COVID-19 pandemic is defining the global health crisis of our time and a major challenge to face. As of May 2021, more than 165 million people have been infected all around the globe; among them, 3.43 million cases of death have been reported to the World Health Organization [1]. With the rapid spread of COVID-19, everything has been impacted; day-to-day life, how we communicate, and how we interact with each other. People are more fearful of crowded spaces, which are seen as a major health risk. In this context, new health measures have been introduced to our daily lives to prevent the infection and reduce the virus' spread, such as avoiding mass gatherings, wearing a mask, and applying social distancing of at least one meter (three ft) between people [2]. Moreover, in large places such as supermarkets and shop stores, strict rules have been applied to keep people safe while they shop, such as limiting the number of people within buildings and adhering to the one meter social distancing rule when queuing. Floor stickers were placed to promote social distancing. However, they are not of much help in monitoring spacing distance.

In recent years, many sectors have benefited from the emergence of wireless networks and the introduction of new connected devices. The Internet of Things is slowly becoming part of every aspect of our lives by giving us more control over simplifying routine work life and complex tasks [3]. IoT is becoming a vital technology in healthcare systems, where it can diagnose, treat, and anticipate risks to patient health [4].

This paper presents a connected object to monitor the spacing distance of one meter between people in queues. The proposed system, based on infrared array sensor MLX90640 to detect people using their temperature and processes it in a low-power microcontroller. In contrast to the most commonly used methods to detect and locate people, such as computer vision technologies, which require high processing capacity to make simultaneous acquisitions and launch human detection algorithms, this node uses a low-resolution thermal image to check the distance between people. The main advantage of this node is that it is not privacy-invasive and consumes less energy compared to camera-based systems. For the communication, we used LoRa technology to send the queue status to the user. This technology was chosen because of its low power consumption, interference immunity, and long-range communication capacity. However, other wireless

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communication technologies could be used to ensure data transfer.

The rest of the paper is organized as follows. In section II, we cite the developed systems for monitoring social distancing. Section III is dedicated to a step-by-step explanation of our original queue management technique, while the object design realization is detailed in section IV. Results and interpretations are given in section V, and finally, a conclusion is provided.

II. Related Works

Recently, many systems have been developed in research and industry sectors to contain and control the spread of the COVID-19 virus. Among them are wearable connected devices that are essentially used to monitor social distancing [5], [6]. These devices are connected to each other and use neighborhood checking processes to alert the user by making vibrations or signal light triggered when the social distancing is no longer respected. Moreover, each device can store details of the interaction between people, which helps to trace the contaminated cases. However, this solution is not adapted to a massive flow of people because it requires many devices, which makes it an expensive tool.

In addition, mobile devices, such as smartphones or smartwatches, are also used to trace the virus, help people make informed decisions about visiting essential locations, and ensure social distancing [7]. As an example, CROWDLESS is a mobile application developed by Ph.D. students of the London School of Economics to indicate the busyness of specific locations in real-time [8]. This application tracks the movements of mobile devices and uses crowd-sourced information reported by users to estimate the people occupying a certain space. Syook [9] is another mobile application that helps to monitor the social distancing in workspaces. It employs embedded Bluetooth devices of smartphones to alert users when the social distancing is not respected. This solution requires having the application installed, and Bluetooth turned on permanently, which drains the battery by keeping the phone busy. Besides, it is up to users to decide whether or not to use the application. Thus, it becomes more challenging to monitor social distancing.

For many years, camera-based systems have been used to track and locate persons using background subtraction or deep neural network processes. These techniques have also been proposed for monitoring social distancing. [10] used a deep learning-based framework to automate the process of monitoring social distancing. Each person is identified as a bounding box using the YOLO v3 object detection model. A Deepsort process is then applied to track and check the social distancing of identified people. This system shows good detection compared to other object detection models such as RCNN and Single Shot Detector (SSD). However, the spatial location of the camera matters and significantly changes the obtained results. Although they have high spatial resolution and good detection, camera-based systems are privacy-invasive, power-consuming, expensive, and highly sensitive to light variations. The following section aims to introduce a new device to monitor social distancing using infrared array sensors.

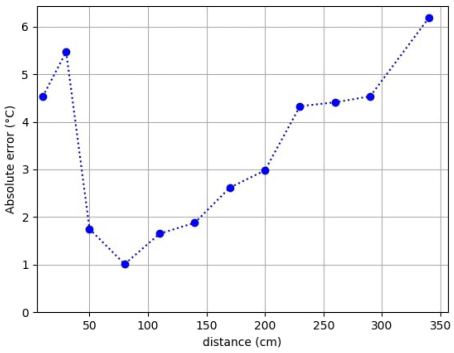


Fig. 1. Absolute error of measured image.

III. Queue Management Technique

The solution that has been developed is based mainly on infrared array sensor MLX90640 [11]. Driven by the low power consumption and low cost, this sensor has been proposed as an effective alternative to camera devices in different applications, such as automatic fall detection and smart security systems [12], [13]. In this work, we use it to detect and check a distance of one meter between people in queues using their body temperature as a metric of processing. Indeed, MLX90640 is a contactless temperature measuring sensor. It provides a low-resolution thermal image composed of 768 pixels (Vertical 24 $\times$ Horizontal 32) using infrared radiation. This sensor has a field of view of 110° $\times$ 75°, where can detect heat sources, and consumes on average 66 mW.

The rest of this section describes the main steps of checking social distancing in queues from a thermal image: the noise removal filter, threshold process, zone detection process, and queue monitoring process.

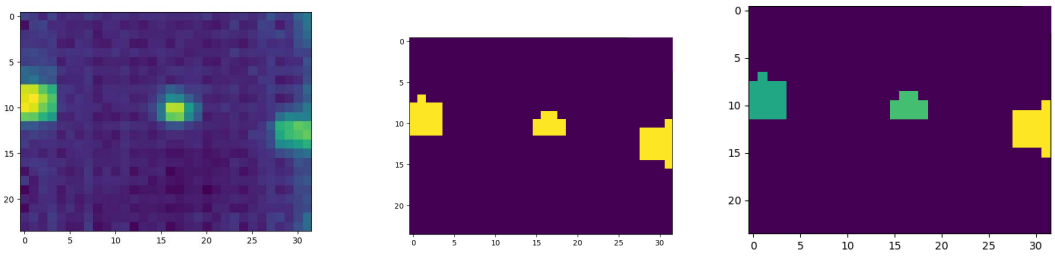
A. MLX90640 Data Analysis

Before introducing the detection algorithm, it will be important to analyze the sensor's raw data. The sensor's sensitivity depends on the field of view and the distance between the target and the sensor. The measured temperature accuracy is not uniform in the image; each pixel has its margin of error, which is low in the center, approximately $\pm$1.5°C, and becomes more significant on the sides. Moreover, the node will be mounted on the roof. Consequently, the height of the room could diminish the accuracy of the measured thermal image. For this reason, we studied the impact of the distance on the measured thermal image (see Fig. 1). The test consisted of measuring an object's temperature (40°C) placed in front of the sensor at different ranges. The room temperature was approximately 23°C, and the refreshing data rate was fixed at 4 Hz. The result shows that the absolute error is high when the object is very close to the sensor or a few meters away. Otherwise, the error of measurements increases slightly from 0.8 m to 2 m, and it is minimal between 0.5 m and 1.5 m (<2°C).

B. Noise Removal Filter

It is more likely that the sensor's raw data present noise when the detected object has a low-temperature value or the

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(a) Filtered data (b) Thresholded data (c) Zone detection result

Fig. 2. People detection process.

reading data rate is high. According to the sensor's datasheet, the root mean square of the error is $+0.5°C$ at 4 Hz and $+1°C$ at 16 Hz. In order to reduce the impact of the noise on the measured thermal image, the Kalman Filter [14] is applied to the sensor's raw data while assuming that the temperature does not vary quickly over time and the unique measured parameter is the temperature. We consider the dynamic system:

$$\begin{cases} x_k = F_k x_{k-1} + u_k \\ y_k = H_k x_k + w_k \end{cases} \quad (1)$$

where x_k is the state variable that represents the temperature of one pixel in the thermal image at time (k) and y_k is the measured temperature of one pixel at time (k). F_k is the system matrix, and H_k is the measurement matrix. u_k and w_k are the state and measurement white noise with covariance Q_k and R_k, respectively. The Kalman filter is composed of two phases: prediction and correction.

In the prediction phase, the state estimate from the previous step $\hat{x}_{k-1|k-1}$ is used to produce the current state estimate $\hat{x}_{k|k-1}$ and the covariance matrix of a prior estimate error in the current step is obtained according to the covariance matrix of the estimated error of system state from the previous step. The prediction equations are expressed as:

$$\hat{x}_{k|k-1} = F_k \hat{x}_{k-1|k-1} \quad (2)$$

$$P_{k|k-1} = F_k P_{k-1|k-1} F_k^\top + Q_k \quad (3)$$

During the correction phase, the current prediction is combined with current measurement information to refine the state estimate. The measurement update equations are given by

$$K_k = P_{k|k-1} H_k^\top (H_k P_{k|k-1} H_k^\top + R_k)^{-1} \quad (4)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k (y_k - H_k \hat{x}_{k|k-1}) \quad (5)$$

$$P_{k|k} = (1 - K_k H_k) P_{k|k-1} \quad (6)$$

where K_k is the Kalman Gain at time (k). $\hat{x}_{k|k}$ and $P_{k|k}$ are, respectively the optimal value of the state estimate (temperature) and the covariance matrix of the estimation error of the system state at time (k). We consider that the process covariance has low value, and we set the covariance noise of the measurements at a calculated value after several acquisitions of the sensor. Finally, we set the refreshing rate at 4 Hz to minimize the presence of the noise.

C. Threshold Process

The purpose of this process is to identify people in the detected area and remove the undesired objects. During this phase, an interval of temperature is calculated $[36.0°C, 39.0°C]$ referring to the distribution of the temperature in the image and the margin error of each pixel as mentioned in Fig. 1 to detect people accurately. This heat interval is applied as a threshold to convert the filtered thermal image into a binary image. Fig. 2b shows the result of this process applied to a captured image of three people standing in a queue (Fig. 2a).

D. Zone Detection Process

At this point, the detection of people becomes easier because each zone in the converted image corresponds to one person standing along the social distancing adhesive tape. In order to identify each zone present in the image, the process starts by attributing to the non-zero value pixel a label after checking whether its neighbors are not labels. Otherwise, the pixel will have the neighbor's label. Finally, the process combines labels by zone using the same neighborhood principal to get an entire label by area. The result of this process in the previous simulation is shown in Fig. 2c.

E. Queue Monitoring Process

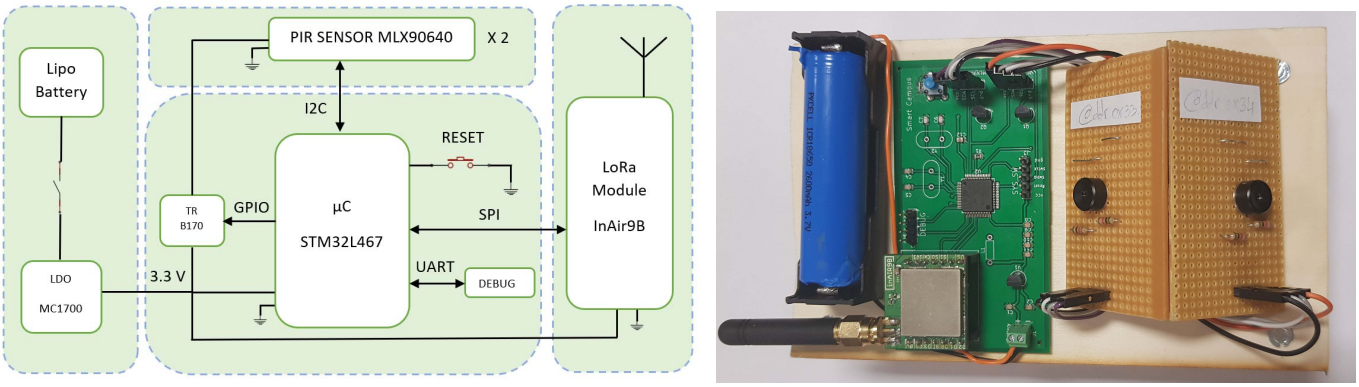
After detecting people in the image, the next step is to check the spacing distance between them in the queue. The position of the sensor matters. Thus, the algorithm used to check social distancing is based on the chosen location and orientation of sensors (Fig. 4). Indeed, the senors are placed to ensure a minimum social distancing of one meter which means that below that range, it is unlikely to have separated zones as obtained in Fig. 2c, where the detected spot takes extra pixels; therefore, being larger than other detected zones allows its identification.

Encouraging results of the algorithm for checking social distancing in queues (see Algorithm 1 in the appendix) made us build the final version of the node considering the constraints of the algorithm and the data acquisition in terms of processing capacity and memory.

IV. Object Design

In terms of hardware, the social distancing node is composed of four elements: the power block, processing block,

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(a) Circuit diagram of the node

(b) Photo of the assembled system

Fig. 3. Social distancing node.

sensing block, and communication block. Fig. 3 shows the overall system block diagram and the assembled circuit board, respectively.

A. Power Block

The power block supplies the main board with 3.3 V. It is composed of a lipo-battery with a capacity of 2.6 Ah and a nominal voltage of 3.7 V and a low dropout voltage regulator (MCP1700) able to deliver up to 250 mA of current to the circuit while consuming only 1.6 μA. Finally, a switch is placed between the battery and the voltage regulator to control the power of the node manually.

B. Processing Block

The processing block, based on the STM32L476 microcontroller to read data from the infrared array sensors, processes it, and sends the information about the social distancing in the queue to the gateways. STM32L476 is part of low-power microcontrollers based on Arm processor Cortex-M4 operating at a frequency up to 80 MHz. It embeds a RAM (random access memory) of 128 kBytes and a Flash memory of 1 MByte.

In this block, the microcontroller uses the I2C protocol to communicate with the senors and the SPI interface to configure and send data to LoRa gateways. In order to conserve the node's energy, the GPIO pins are used to power on the sensors only when necessary. Finally, the UART protocol is used for debugging the code.

C. Sensing Block

As shown in Fig. 3b, this block is composed of two MLX90640 infrared array sensors oriented in opposite directions to detect the maximum possible of people standing in the queue. The measured images will be converted into one thermal image before being processed in the algorithm for checking social distancing. Moreover, the node is scalable. It is possible to add more sensors to the node in the same bus of communication, I2C, to enlarge the field of detection.

D. Communication Block

1) Wireless Communication Technologies: Over the last years, several communication protocols and technologies have been designed for IoT applications. The most popular low-power technologies are Bluetooth Low Energy (BLE), Zigbee, and Low Power Wide Area Network (LPWAN), such as Sigfox and LoRa [15], [16]. A benchmark of technologies is provided in Table I. Each technology has its advantages and limitations. For example, BLE and Zigbee provide a high data rate compared to the LPWAN technologies. However, LoRa and Sigfox technologies can communicate at long ranges. Therefore, depending on the application, the technology can be chosen considering its power consumption, range, topology, and communication rate.

TABLE I
WIRELESS COMMUNICATION TECHNOLOGIES

	BLE	Zigbee	Sigfox	LoRa
Bandwidth	2 MHz	1 MHz	100 Hz	250 kHz
Modulation	GFSK	DSSS	DBPSK	CSS
Topology	Meshed	Meshed	Star	Star
PHY rate	125-500 kb/s	250 kb/s	0.1 - 1 kb/s	0.25 - 11 kb/s
Range	10 m	100 m	up to 40 Km	up to 10 Km

In this work, we are interested in star networks and long-range technologies, which are more appropriate to places with large surfaces, such as campuses and supermarkets. The LoRa technology is chosen because of its long-range communication capacity, interference immunity, and the possibility of deploying a private network. This technology is based on chirp spread spectrum (CSS) modulation, in which a specific parameter called Spreading Factor (SF) is used to control the bit rate, the range, and the energy consumption [17]. LoRa devices communicate over the unlicensed radio spectrum in Industrial, Scientific, and Medical (ISM) bands over 169, 433, 868 MHz (in Europe) and 915 MHz and respects the regulations that fix the duty cycle (ranging from 0.1 to 1% depending on the used sub-band) and the transmission power.

2) LoRa Module: The InAir9B LoRa module is used to send the status of the queue (respecting or not respecting social distancing) to the gateways. This module is based on the Semtech SX1276 transceiver with a standard SMA connector and SPI interface. The module communicates over 868 MHz bands and has a maximum output power of +20 dBm. The communication between the node and LoRa gateways is

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TABLE II
LoRa COMMUNICATION PARAMETERS

Parameter	value
Frequency	868 MHz
Spreading Factor	12
Coding rate	4/5
Bandwidth	125 KHz
Transmission power	13 dBm

TABLE III
THE COST OF THE NODE COMPONENTS

Component	Approximate Cost
Lipo-Battery	10.98 €
Microcontroller	4.49 €
PIR sensor (MLX90640) x 2	74.86 €
LoRa module	6.25 €
LDO	0.34 €
Transistor BS170 x 2	0.5 €
PCB manufacturing	2 €
Total Cost	99.42 €

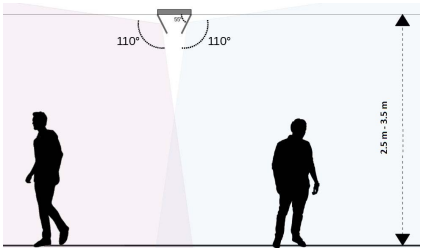


Fig. 4. The position of the node.



Fig. 5. The configuration of the first experiment.

TABLE IV
THE LOCATION AND THE HEIGHT OF PARTICIPANTS

Location	1st	2nd	3th	4th	5th	6th
Height	1.71 m	1.69 m	1.75 m	1.78 m	1.77 m	1.72 m

TABLE V
RESULTS OF THE FIRST EXPERIMENT

Height	2.5 m	2.8 m	3.0 m	3.2 m	3.5 m
Error (e)	27 %	8 %	22 %	36 %	52 %

carried out using only the physical layer of communication (LoRa modulation) to conserve energy. The main parameters of transmission are listed in Table II.

E. The Cost of the Node

The breakdown of cost for the parts required to manufacture the social distancing node is shown in Table III.

The social distancing node can be constructed for approximately 100 €. This price is still cheaper than the camera-based systems, which require high embedded processing boards and accurate measurements of cameras. However, it should be noted that such expensive systems are expected to be more accurate than the proposed solution.

V. RESULTS AND DISCUSSION

A. The Node Results

The position and direction of sensors are important characteristics to locate people precisely and check the distance between them. The PIR sensors are placed at an angle of 55° from the horizontal view (Fig. 4). The inclination angle is calculated referring to the field of view of sensors to ensure a full view of the queue and good matching of the measured images. The first experiment consisted of forming a queue of six persons respecting social distancing of one meter (Fig. 5) and one node mounted on the roof of the room and located at the center of the queue. Table IV shows the height of the participants and their location in the queue. To evaluate the system's performance, we fixed the node at different heights between 2.5 meters and 3.5 meters from the ground. This experiment has been conducted at room temperature, 24°C.

Table V summarizes the performance of the node at a different height. Considering the setting of the experiment, the error of checking the social distancing between participants at each configuration was calculated from the false alerts of the node using the following equation:

$$e = \frac{\sum_{iter=1}^{N} (F_a)_{iter}}{N} \times 100 \tag{7}$$

where N is the total number of measurements and F_a is a number equal to 1 if the node make a false alert (detecting a violation of social distancing) and 0 otherwise.

The result was obtained after 1,000 measurements at each configuration. As shown in Table V, the error of detection is low at 2.8 meters. In this configuration, each sensor can detect and locate up to four persons in its field of view. The processed data in Fig. 6 illustrate a perfect separation of our participants and clear detection of zones. However, the performance deteriorated at 2.5 meters of height because the node was close to the participants. Severe interference was created in the middle of the queue, more precisely, between the 3rd and 4th participants. Above three meters, the system can detect up to nine persons in the queue, but the resolution limitation of the sensors strongly impacts their detection accuracy. It was noticed that two or more persons are captured as one heat block at the beginning and the end of

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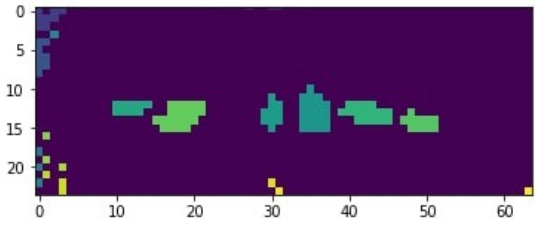


Fig. 6. Detected image at 2.8 m of height.

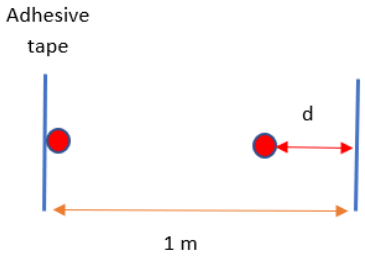


Fig. 7. The setting of the second experiment.

TABLE VI
RESULT OF THE SECOND EXPERIMENTATION

Distance d	5 cm	10 cm	15 cm	20 cm	25 cm or more
Precision (p)	32 %	54 %	76 %	93 %	100 %

the queue. Therefore, excluding the configuration of 2.8 meters, the checking distance process does not reflect the reality of the scenario.

Since the configuration of 2.8 meters conducted good detection, we focused on this configuration. We then conducted another experiment in the case of not respecting social distancing between participants. This experiment consisted of making a queue of two persons in which the first participant respected the limit fixed by the adhesive tape and the second participant crossed the line by a distance d (Fig. 7). The participants' heights were, respectively, 1.75 meters and 1.78 meters, and the room temperature was 24°C.

Table VI shows the results of the experiment after 1,000 measurements at different values of distance d between the second participant and the line. The precision of checking the distance (Equation 8) between the participants is calculated using the missed detection of the node (not detecting the violation of social distancing).

$$p = (1 - \frac{\sum_{iter=1}^{N} (M_d)_{iter}}{N}) \times 100 \quad (8)$$

N is the total number of measurements, and M_d is a number equal to 1 if the node is not detecting a violation of social distancing and 0 otherwise. In this experiment, the node successfully detected violations of social distancing (93%) when the distance between participants was below 80 cm. Between 80 cm and 95 cm, the accuracy of the node has deteriorated. Therefore, for a rule of social distancing of one meter, the node has a margin of error of 20 cm.

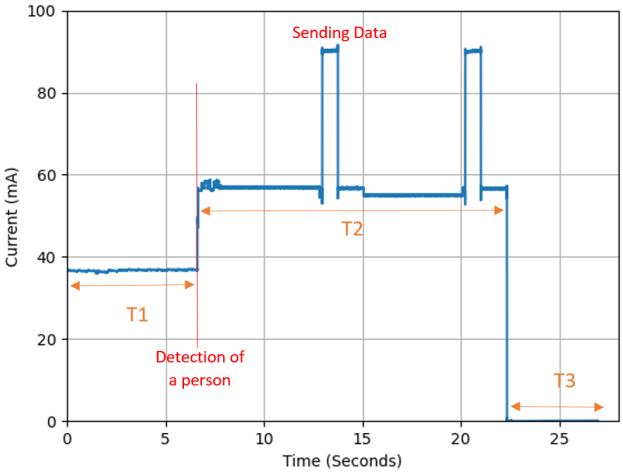


Fig. 8. Current consumption of the node.

Nevertheless, the obtained results of the node could be impacted in particular situations because of the sensor limitations and settings used for checking social distancing:

- High temperature of the environment: It will be more difficult to accurately detect people forming the queue based on their body temperature.
- Covered people: The detection of the node is based mainly on people's heat information, particularly on the top side of their bodies. The MLX90640 sensor provides a thermal image using infrared radiation. Thus, it will be more difficult to detect people if they are covered.
- Hiding persons in a queue: A tall person could hide short people behind him because of the inclination of the sensor. This problem can be resolved by separating the PIR sensors of the node by a distance of 2.5 meters and orienting them toward the ground (0° from the horizontal view). However, the capacity of the node will be reduced to four persons per queue.

The node's main advantage is its ability to preserve people's privacy while checking social distancing, all by doing very little data processing and providing a real-time response. However, the system is limited to six persons per queue. Thus, to enlarge the monitoring capacity, more nodes should be mounted at the same queue.

B. Power Consumption

The power consumption of the node is analyzed dynamically through the Joulescope board [18]. Fig. 8 describes the current consumption of the node at different steps:

- T1: Reading data from one sensor (the queue is empty).
- T2: Checking social distancing and sending data.
- T3: Sleep mode.

According to Fig. 8, the node consumes on average 57 mA in run mode (T2) and less than 12 μA in sleep mode (T3). During run mode, each sensor consumes on average 20 mA, the lora module consumes 30 mA, and the microcontroller consumes approximately 14 mA because it operates at the maximum frequency of processing (80 MHz). This frequency is optimal for the system, and it was chosen after comparing the energy consumption of the node using different frequencies.

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Algorithm 1 Algorithm for Checking Social Distancing

Data: $T[64][24]$ Filtered image, D_{max} max density of zone
Result: $result$ state of the queue
Initialisation: $B[64][24]$ Binary image, $label \leftarrow 0$, $Label[64][24]$ label table, $Density[20]$ density of labels in image, $result \leftarrow 0$;
for $i = 0$ *to* 64 **do**
  for $j = 0$ *to* 24 **do**
    if $T[i, j] \in [36.0°C, 39.0°C]$ **then**
      $B[i, j] \leftarrow 1$
    else
      $B[i, j] \leftarrow 0$
    end
  end
end
for $i = 0$ *to* 64 **do**
  for $j = 0$ *to* 24 **do**
    if $B[i, j] \neq 0$ **then**
      if $B[neighbor(i, j)] \neq 0$ **then**
        $Label[i, j] \leftarrow Label[neighbor(i, j)]$
      else
        $Label[i, j] \leftarrow label + 1$
      end
    end
  end
end
for $i = 0$ *to* 64 **do**
  for $j = 0$ *to* 24 **do**
    if $(Label[i, j] \times Label[neighbor(i, j)] \neq 0)$ *and* $(Label[i, j] \neq Label[neighbor(i, j)])$ **then**
      change all $Label[i, j]$ value in $Label$ by $Label[neighbor(i, j)]$
    end
  end
end
$k \leftarrow 0$;
for $i = 0$ *to* 64 **do**
  for $j = 0$ *to* 24 **do**
    if $Label[i, j] \neq 0$ **then**
      Density[k] = count $Label[i, j]$ in $Label$
      change all $Label[i, j]$ value in $Label$ by 0
      $k \leftarrow k + 1$;
    end
  end
end
for $i = 0$ *to* k **do**
  if $Density[i] \geq D_{max}$ **then**
    $result \leftarrow result + 1$
  end
end

Since power consumption is of much interest, we have been looking for optimal methods to minimize the node's power consumption. Analyses reveal that reading data from the two sensors and sending data are the most energy-consuming steps. Thus, we have decided to use the second sensor only when needed; it stays in sleep mode until a person is detected in the queue (T1). In addition, the node will send data only when the social distancing is no longer respected.

VI. Conclusion

This paper presents a real-time queue monitoring system based on infrared technology. It uses a data processing algorithm to locate persons in a crowded space by detecting each zone in the IR array. The proposed method is used to monitor the spacing distance between people in queues to avoid the spread of the COVID-19 virus, as recommended by the World Health Organization. The main advantage of this system is its ability to respect people's privacy while boating a lower power consumption. Compared to other systems based on cameras, this one requires little data processing and no need for data transfer. It has been successfully used to monitor social distancing in indoor environments and achieved our goals of real-time performance. The proposed hardware is portable, low-cost, and easy to set up. The system operates on STM32L476 and uses LoRa technology for communication. In the following weeks, the node will be deployed at the university campus to ensure social distancing. However, this does not preclude using the node for other applications, such as managing the occupation of rooms.

Appendix

See Algorithm 1.

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